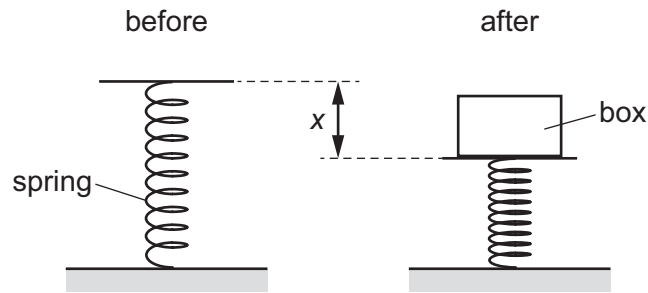


- 21** A platform is supported by a spring. A box is placed onto the platform. The diagram shows the spring before and after the box is placed onto it.



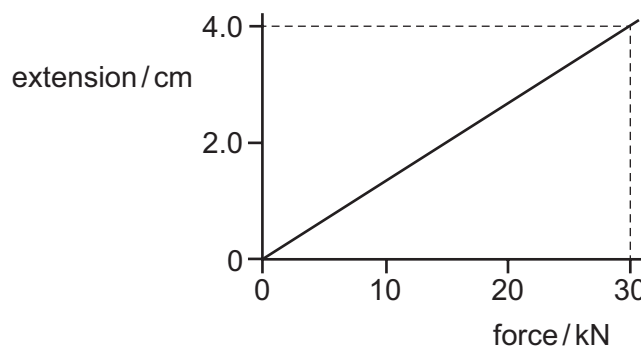
What is the term for the quantity marked as x ?

- A** compression
 - B** load
 - C** strain
 - D** stress
- 22** A copper wire has length 1.7 m and uniform diameter 0.64 mm.

The Young modulus of copper is 1.2×10^{11} Pa.

What is the spring constant of the wire?

- A** $2.3 \times 10^4 \text{ N m}^{-1}$
 - B** $9.1 \times 10^4 \text{ N m}^{-1}$
 - C** $4.5 \times 10^7 \text{ N m}^{-1}$
 - D** $7.1 \times 10^7 \text{ N m}^{-1}$
- 23** The graph shows how the extension of a spring varies with the force used to stretch it.



What is the elastic potential energy of the spring when the extension is 4.0 cm?

- A** 60 J
- B** 120 J
- C** 600 J
- D** 1200 J